Date:28/07/2026

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("/content/diabetes1.csv")
```

```
df.columns
```

```
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Diabetes'],
      dtype='object')
```

```
df.shape
```

```
(768, 9)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Pregnancies               768 non-null    int64  
 1   Glucose                   768 non-null    int64  
 2   BloodPressure             768 non-null    int64  
 3   SkinThickness             768 non-null    int64  
 4   Insulin                   768 non-null    int64  
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64  
 8   Diabetes                  768 non-null    int64  
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

```
df.describe()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Diabetes
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	0.471876	33.240885	0.348958
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	21.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.243750	24.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	41.000000	1.000000
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	2.420000	81.000000	1.000000

```
df.isnull().sum()
```

	0
Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Diabetes	0

dtype: int64

```
df["Diabetes"].value_counts()
```

Diabetes	count
0	500
1	268

dtype: int64

```
cols = ["Glucose","BloodPressure","SkinThickness","Insulin","BMI"]
(df[cols] == 0).sum()
```

	0
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11

dtype: int64

```
import numpy as np
cols = ["Glucose","BloodPressure","SkinThickness","Insulin","BMI"]
df[cols] = df[cols].replace(0,np.nan)
```

```
df
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Diabetes
0	6	148.0	72.0	35.0	NaN	33.6	0.627	50	1
1	1	85.0	66.0	29.0	NaN	26.6	0.351	31	0
2	8	183.0	64.0	NaN	NaN	23.3	0.672	32	1
3	1	89.0	66.0	23.0	94.0	28.1	0.167	21	0
4	0	137.0	40.0	35.0	168.0	43.1	2.288	33	1
...	...	...	...	...	...	...	...	...	...
763	10	101.0	76.0	48.0	180.0	32.9	0.171	63	0
764	2	122.0	70.0	27.0	NaN	36.8	0.340	27	0
765	5	121.0	72.0	23.0	112.0	26.2	0.245	30	0
766	1	126.0	60.0	NaN	NaN	30.1	0.349	47	1
767	1	93.0	70.0	31.0	NaN	30.4	0.315	23	0

768 rows × 9 columns

```
df.tail(60)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Diabetes
708	9	164.0	78.0	NaN	NaN	32.8	0.148	45	1
709	2	93.0	64.0	32.0	160.0	38.0	0.674	23	1
710	3	158.0	64.0	13.0	387.0	31.2	0.295	24	0
711	5	126.0	78.0	27.0	22.0	29.6	0.439	40	0
712	10	129.0	62.0	36.0	NaN	41.2	0.441	38	1
713	0	134.0	58.0	20.0	291.0	26.4	0.352	21	0
714	3	102.0	74.0	NaN	NaN	29.5	0.121	32	0
715	7	187.0	50.0	33.0	392.0	33.9	0.826	34	1
716	3	173.0	78.0	39.0	185.0	33.8	0.970	31	1
717	10	94.0	72.0	18.0	NaN	23.1	0.595	56	0
718	1	108.0	60.0	46.0	178.0	35.5	0.415	24	0
719	5	97.0	76.0	27.0	NaN	35.6	0.378	52	1
720	4	83.0	86.0	19.0	NaN	29.3	0.317	34	0
721	1	114.0	66.0	36.0	200.0	38.1	0.289	21	0
722	1	149.0	68.0	29.0	127.0	29.3	0.349	42	1
723	5	117.0	86.0	30.0	105.0	39.1	0.251	42	0
724	1	111.0	94.0	NaN	NaN	32.8	0.265	45	0
725	4	112.0	78.0	40.0	NaN	39.4	0.236	38	0
726	1	116.0	78.0	29.0	180.0	36.1	0.496	25	0
727	0	141.0	84.0	26.0	NaN	32.4	0.433	22	0
728	2	175.0	88.0	NaN	NaN	22.9	0.326	22	0
729	2	92.0	52.0	NaN	NaN	30.1	0.141	22	0
730	3	130.0	78.0	23.0	79.0	28.4	0.323	34	1
731	8	120.0	86.0	NaN	NaN	28.4	0.259	22	1
732	2	174.0	88.0	37.0	120.0	44.5	0.646	24	1
733	2	106.0	56.0	27.0	165.0	29.0	0.426	22	0
734	2	105.0	75.0	NaN	NaN	23.3	0.560	53	0
735	4	95.0	60.0	32.0	NaN	35.4	0.284	28	0
736	0	126.0	86.0	27.0	120.0	27.4	0.515	21	0

```
df
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Diabetes
0	6	148.0	72.0	35.0	NaN	33.6	0.627	50	1
1	1	85.0	66.0	29.0	NaN	26.6	0.351	31	0
2	8	183.0	64.0	NaN	NaN	23.3	0.672	32	1
3	1	89.0	66.0	23.0	94.0	28.1	0.167	21	0
4	0	137.0	40.0	35.0	168.0	43.1	2.288	33	1
5	5	116.0	74.0	NaN	NaN	25.6	0.201	30	0
6	3	78.0	50.0	32.0	88.0	31.0	0.248	26	1
7	10	115.0	72.0	NaN	NaN	35.3	0.134	29	0
8	2	197.0	70.0	45.0	543.0	30.5	0.158	53	1
9	8	125.0	96.0	NaN	NaN	0.0	0.232	54	1
10	4	110.0	92.0	NaN	NaN	37.6	0.191	30	0
11	10	168.0	74.0	NaN	NaN	38.0	0.537	34	1
12	10	139.0	80.0	NaN	NaN	27.1	1.441	57	0
13	1	189.0	60.0	23.0	846.0	30.1	0.398	59	1
14	5	166.0	72.0	19.0	175.0	25.8	0.587	51	1
15	7	100.0	84.0	33.0	105.0	30.0	0.488	46	0
16	0	118.0	84.0	47.0	230.0	45.8	0.551	31	1
17	7	107.0	74.0	NaN	NaN	29.6	0.254	31	1
18	1	103.0	30.0	38.0	83.0	43.3	0.183	33	0
19	1	115.0	70.0	30.0	96.0	34.6	0.529	32	1

768 rows × 9 columns

```
df.isnull().sum()
```

	0
Pregnancies	0
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11
DiabetesPedigreeFunction	0
Age	0
Diabetes	0

dtype: int64

```
cols = ["Glucose","BloodPressure","SkinThickness","Insulin","BMI"]
for col in cols:
 df[col] = df[col].fillna(df[col].median())
```

```
df.isnull().sum()
```

	0
Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Diabetes	0

dtype: int64

```
df.head()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Diabetes
0	6	148.0	72.0	35.0	125.0	33.6	0.627	50	1
1	1	85.0	66.0	29.0	125.0	26.6	0.351	31	0
2	8	183.0	64.0	29.0	125.0	23.3	0.672	32	1
3	1	89.0	66.0	23.0	94.0	28.1	0.167	21	0
4	0	137.0	40.0	35.0	168.0	43.1	2.288	33	1

```
df.tail()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Diabetes
763	10	101.0	76.0	48.0	180.0	32.9	0.171	63	0
764	2	122.0	70.0	27.0	125.0	36.8	0.340	27	0
765	5	121.0	72.0	23.0	112.0	26.2	0.245	30	0
766	1	126.0	60.0	29.0	125.0	30.1	0.349	47	1
767	1	93.0	70.0	31.0	125.0	30.4	0.315	23	0

date: 04/08/2026

```
# --> single line comment
```

```
print("This is our  data sccience class \n")
print("this is our first code" )
```

```
This is our  data sccience class

this is our first code
```

```
# variable
a=12
b=3.5
c="Data Science"
d=(a+b)
print(d)
print(a)
print(type(a))
print(id(a))
print(type(b))
print(id(b))
print(type(c))
print(id(c))
```

```
15.5
12
<class 'int'>
11645704
<class 'float'>
133557739088752
<class 'str'>
133557737709232
```

```
#multiple access
a,b,c=50,30,60
print(a+b+c)

a=b=c=90
print(a)
print(a+b+c)

a,b,name=50,20.25,"abc"
print(a)
print(b)
print(name)
print(a,"\n",b,"\n",name)
```

```
140
90
270
50
20.25
abc
50
 20.25
 abc
```

```
# type casting
a=(int(input("enter the first number:")))
b=(int(input("enter the second number:")))
print("sum of two number is :",a+b)
print("Difference of two number is :",a-b)
print("Product of two number is :",a*b)
print("division of two number is :",a/b)
print("Quotient of two number is :",a%b)
```

```
enter the first number:40
enter the second number:70
sum of two number is : 110
Difference of two number is : -30
Product of two number is : 2800
division of two number is : 0.5714285714285714
Quotient of two number is : 40
```

```
# conditional statement
# if  ,  else  , elif  ,  nested if

age = (int(input("enter the Age:")))
if age>=18:
  print("you are eligible for voting")
else:
  print("you are not eligible for voting")
```

```
enter the Age:40
you are eligible for voting
```

```
# if else

num = (int(input("enter the number:")))
if age>0:
  print("positive number")
else:
  print("negative number")
```

```
enter the number:10
positive number
```

```
# positive number , negative number , zero number
num = (int(input("enter the number:")))
if num==0:
  print("this number is zero number")
elif num>=1:
  print("this number is positive number")
else:
  print("this number is negative number")
```

```
enter the number:0
this number is zero number
```

```
# even number , odd number
num = (int(input("enter the number:")))
if num%2==0:
  print("this number is even number")
else:
  print("this number is odd number")
```

```
enter the number:12
this number is even number
```

```
# find the largest and smallest number
num1 = (int(input("enter the number:")))
num2 = (int(input("enter the number:")))
num3 = (int(input("enter the number:")))
if num1>num2 and num1>num3:
  print("num1 is largest number")
elif num2>num1 and num2>num3:
  print("num2 is largest number")
else:
  print("num3 is largest number")

if num1<num2 and num1<num3:
  print("num1 is smallest number")
elif num2<num1 and num2<num3:
  print("num2 is smallest number")
else:
  print("num3 is smallest number")

print("the largest number is :", max(num1,num2,num3))
print("the smallest number is :", min(num1,num2,num3))
```

```
enter the number:12
enter the number:18
enter the number:40
num3 is largest number
num1 is smallest number
the largest number is : 40
the smallest number is : 12
```

```
# to check whether the number is leap year or not
num = (int(input("enter the number:")))
if (num % 4 == 0 and num % 100 !=0) or (num % 400 == 0):
  print("this number is leap year")
else:
  print("this number is not leap year")
```

```
enter the number:2000
this number is leap year
```

```
count = 1
while count <= 10:
  print(count)
  count+=1
```

```
1
2
3
4
5
6
7
8
9
10
```

```
# write a pythone program to reverse a number
num = int(input("enter the number :"))
d=0
while num/10:
  d=d*10+num%10
  num=num//10
print(d)
```

```
enter the number :12
21
```

```
# write a progeram whether a number is palindrome
num = int(input("enter the number :"))
a=num
d=0
while num/10:
  d=d*10+num%10
  num=num//10
print(d)
if a==d:
  print("it is a palindrome")
```

```
else:
  print("not palindrome")
```

```
enter the number :121
121
it is a palindrome
```

```
# range (stop)
# range(start , stop)
# range(start , stop , step)
# for loop

for i in range(1 , 6):
  print(i)

for j in range(1 , 10 , 1):
  print(j)
```

```
1
2
3
4
5
1
2
3
4
5
6
7
8
9
```

DATE : 11/08/2026

```
#string function
str_name="Aayush kumar Choubey"
print(type(str_name))
print(str_name.upper())
print(str_name.lower())
print(str_name.title())
print(str_name.capitalize())
print(str_name[4])
print(str_name[-1])
print(str_name[-5])
```

```
<class 'str'>
AAYUSH KUMAR CHOUBEY
aayush kumar choubey
Aayush Kumar Choubey
```

```
Aayush kumar choubey
s
y
o
```

```
#concatinate
F_name="Aayush"
L_name="kumar"
full_name=F_name+L_name
print(full_name)
# double cort for the space between the name
full_name=F_name+" "+L_name
print(full_name)
```

```
Aayushkumar
Aayush kumar
```

```
#concatinate
a = "Aayush kumar"
b = 7
str1 = a+str(b)
print(str1)
```

```
Aayush kumar7
```

```
#string sclicing
a="Aayush kumar choubey"
print(a[3:6])
print(a[:5])
print(a[1:-3])
print(a[5:])
print(a[-2:-1])
```

```
ush
Aayus
ayush kumar chou
h kumar choubey
e
```

```
a=[1,2]
b=a
c=[1,2,3]
print(a is not c)
print(3 in c)
```

```
True
True
```

```
a="python programming class is good  "
print(a.title())
print(a.capitalize())
print(a.strip())
print(a.lstrip())
print(a.rstrip())
print(a.replace("python","java"))
#  print(a.find())
#  print(a.count())
#  print(a.startswith())
```

```
Python Programming Class Is Good
Python programming class is good
python programming class is good
python programming class is good
python programming class is good
java programming class is good
```

```
#         LIST

list1=[1,2,3,4,5]
list2=[1,"abc","a",2.25,2]
list3=[25,"abcd"]
print(list1)
print(list2)
print(list3)
print(list1[1:3])
print(list2[:-1])
print(list3[0:1])
print(list1*2)
print(list2*3)
print(list1+list2+list3)
```

```
[1, 2, 3, 4, 5]
[1, 'abc', 'a', 2.25, 2]
[25, 'abcd']
[2, 3]
[1, 'abc', 'a', 2.25]
[25]
[1, 2, 3, 4, 5, 1, 2, 3, 4, 5]
[1, 'abc', 'a', 2.25, 2, 1, 'abc', 'a', 2.25, 2, 1, 'abc', 'a', 2.25, 2]
[1, 2, 3, 4, 5, 1, 'abc', 'a', 2.25, 2, 25, 'abcd']
```

```
#                 Tuples
tup1=("a","abc",25,2.25)
tup2=("xyz",95)

print(tup1)
print(tup2)
print(tup1[3])
```

```
print(tup1[2:4])
print(tup1*3)
print(tup1+tup2)
```

```
('a', 'abc', 25, 2.25)
('xyz', 95)
2.25
(25, 2.25)
('a', 'abc', 25, 2.25, 'a', 'abc', 25, 2.25, 'a', 'abc', 25, 2.25)
('a', 'abc', 25, 2.25, 'xyz', 95)
```

```
#           Dictionary
dict={"item":"Apple","price":200,"color":"red",}
print(dict)
print(dict["item"])
print(dict["price"])
print(dict["color"])
del dict["price"] # used to delete the value in dictionary
dict["Amount"]="1 kg" # used to add the value in dictionary
print(dict["Amount"])
print(dict)
```

```
{'item': 'Apple', 'price': 200, 'color': 'red'}
Apple
200
red
1 kg
{'item': 'Apple', 'color': 'red', 'Amount': '1 kg'}
```

```
#                              SETS
s1={"apple","banana","orange"}
s2={2,3,4,5,6,7,8,9,15,16,17}
s3={1,9,10,11,12,13}
print(s2,"\n",s3)

#union(|)
print(s2|(s3))
print(s2.union(s3))

#intersection(&)
print(s2&(s3))
print(s2.intersection(s3))

#difference(-)
print(s2-(s3))
print(s2.difference(s3))

#symmetric Difference(^)
print(s2^(s3))
print(s2.symmetric_difference(s3))
```

```
#add
s2.add(45)
print(s2)

#remove
s2.remove(15)
print(s2)

#clear
s2.clear()
print(s2)


#discard
s2.discard(10)
print(s2)
```

```
{2, 3, 4, 5, 6, 7, 8, 9, 15, 16, 17}
 {1, 9, 10, 11, 12, 13}
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17}
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17}
{9}
{9}
{2, 3, 4, 5, 6, 7, 8, 15, 16, 17}
{2, 3, 4, 5, 6, 7, 8, 15, 16, 17}
{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 15, 16, 17}
{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 15, 16, 17}
{2, 3, 4, 5, 6, 7, 8, 9, 45, 15, 16, 17}
{2, 3, 4, 5, 6, 7, 8, 9, 45, 16, 17}
set()
set()
```

```
#write a python program to find the sum of all element in a list
a=[1,2,3,4,5,6,7,8,9,10]
sum = 0
for n in range(0,10):
  sum+=(a[n])
print(sum)
```

```
55
```

```
#write a python program to find the sum of all element in a list


# a = (input("Enter values :").split())
# sum = 0
# for n in range(0,10):
#   sum+=(a[n])
# print(sum)
```